

RetailCo Data Platform

Brief Business Insights

Team D | Stage 8 | HNG Internship | May 2026

RetailCo operates four stores across Lagos, Abuja, Port Harcourt and Kano. All insights below are drawn from 132,559 records loaded into a PostgreSQL warehouse and queried through dbt analytical models. No estimates or assumptions are used. Every finding reflects a live SQL query.

Q1. Revenue Performance by Store

The result here was a surprise. Kano came out top in both total revenue and transaction count, not Lagos. Lagos ranked second by transaction count. Abuja recorded the highest average order value per transaction.

Population density does not automatically translate to retail revenue. Kano likely benefits from less competition in the area, bulk purchasing patterns common in northern Nigerian markets, and a product mix suited to local demand. Abuja's high average order value reflects FCT purchasing power even at lower purchase frequency.

Recommendation: Study what the Kano store is doing differently and apply those learnings to the other locations before assuming Lagos needs more investment.

Q2. Customer Behaviour and Segmentation

Premium segment customers purchase less often but spend more per visit. Mass market customers come in more frequently with smaller baskets. The highest value group across the dataset is customers who made three or more purchases within any 90-day window. These customers show strong loyalty and represent the best return on any retention spend.

The dim_customer table uses SCD2 tracking, so when a customer moves between segments, both the old and new records are preserved. Historical revenue attribution stays accurate regardless of when the segment change happened.

Recommendation: Build a loyalty programme targeting the 3-plus purchase cohort. Early access to new products or a basic points scheme could significantly improve retention at low cost.

Q3. Product Performance and Discount Analysis

The most pressing finding in this section is products with thin margins being discounted on top of that. When both conditions exist together, the store is selling at a loss without flagging it. The analysis identified several products in this position across the four stores.

The dim_product table also uses SCD2. When a product price changes, all historical sales still reflect the price that was active at the time of the original transaction. This keeps trend analysis reliable over time.

Recommendation: Any discount above 20 percent on a product with a margin below 15 percent should require manager approval. This one rule, applied consistently, could recover meaningful revenue without any change to pricing strategy.

Q4. Payment Channel Insights

Mobile money is the dominant payment method in Lagos and Port Harcourt. Card payments are stronger in Abuja, which reflects higher banking penetration in the FCT. The payment data also surfaced 566 records with zero amounts or unexplained negative values.

These 566 payments were isolated into a dedicated `flagged_payments` table and removed from all revenue calculations. The most probable cause is a POS integration error rather than fraud, but they need investigation within 30 days to confirm.

Recommendation: Investigate the flagged payments and resolve the root cause. Consider expanding mobile money acceptance in Kano to reduce cash dependency at that location.

Q5. Operational Data Quality

Three quality monitors run automatically every night as part of the dbt pipeline. Orders stuck in pending status for more than seven days are flagged for store operations. Products with no inventory movement in 30 days are flagged for a stock review. Zero-amount payments are captured separately for finance to investigate.

None of these require manual checking. The results are always available as a live query in the warehouse. The only thing required from the business side is assigning someone responsible for acting on each output.

Recommendation: Assign ownership of each monitor to a named team member with a clear response window. Data quality outputs are only useful when someone is accountable for acting on them.

Summary

- **Revenue:** Kano leads in total revenue and transactions. Abuja leads in average order value. Lagos is second by volume.
- **Customers:** The highest value group makes three or more purchases in 90 days. Prioritise retention for this group.
- **Products:** Discounting low-margin products is the single biggest profitability risk. Introduce an approval policy.
- **Payments:** Mobile money dominates in Lagos and Port Harcourt. 566 anomalous payments need investigation.
- **Data Quality:** Three monitors are running automatically. Assign ownership and act on the outputs weekly.

GitHub: github.com/Yusinho/retailco-pipeline

Demo: <https://www.loom.com/share/2547e57635754767ac1def319d654d23>